Article

Predicting Football Match Outcomes Using Machine Learning

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Abstract

Football match outcomes are influenced by a complex interplay of dynamic team strategies and stochastic match events, posing a significant challenge for predictive analytics. In this paper, we propose a unified benchmarking and calibration framework for football forecasting, utilizing a dataset of approximately 230,000 matches. We systematically compare diverse learning paradigms—ranging from linear models (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree, LightGBM, Random Forest) and Neural Network models (MLP). To support future probability calibration, bookmaker odds are transformed using the Newton–Raphson method to remove bookmaker margins. The resulting predictive framework establishes the basis for the subsequent integration of an Isotonic Calibration layer for probability estimation and betting signal generation. Our empirical results demonstrate that while methods like Random Forest excel in overall Precision (reaching 65.11% for Home Win and 69.07% for Away Win), the Multi-Layer Perceptron (MLP) yields superior non-linear feature interaction mapping, achieving the highest Recall across both markets (63.94% and 60.25% respectively). Our findings show that advanced feature engineering combined with systematic benchmarking enables the development of reliable predictive models, while establishing the foundation for subsequent probability calibration and betting-oriented decision support.

Keywords: Machine learning; neural networks; stochastic techniques.

1. Introduction

Many real world problems can be tackled by the so - called machine learning tools. Among these problems one can find cases derived from various scientific fields, such as physics [1,2], astronomy [3,4], chemistry [5,6], medicine [7,8], economics [9,10], image processing [11], time series forecasting [12] etc. Common machine learning tools are artificial neural networks [13,14], decision trees [15,16], support vector machines (SVM) [17,18] etc. In this research study, we address the problem of predicting the outcome of a football match by examining it through a variety of machine learning techniques. This problem was tackled by various researches during the past years.

Predicting football match outcomes is difficult because of the high levels of noise, changing team strategies, and unpredictable match events. Standard predictive models often struggle in these volatile environments. This research is important for three main reasons: First, it addresses the core issue of probability calibration in sports analytics. Machine learning models often produce raw scores that do not reflect real-world probabilities,

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especially when the data is noisy. By implementing methods like Isotonic Calibration and using the Newton–Raphson [19] approach to remove bookmaker margins, we provide a framework that transforms biased model outputs into reliable, objective probabilities. Second, we conduct a rigorous comparative analysis under strictly uniform conditions. Instead of focusing on just one algorithm, we test fundamentally different models—including linear models, GAM [20], tree structures (FastTree [21], LightGBM [22], Random Forest [23]), and Neural Network models (MLP)—using the exact same input features. This provides an unbiased look at how each model handles the non-linearity and volatility of sports data. Finally, from a practical standpoint, this research highlights the important trade-off between Precision and Recall in different betting markets. Whether we are analyzing Home Win and Away Win markets, finding the right balance is essential. Our results show that the Multi-Layer Perceptron (MLP), when integrated into our calibrated framework, consistently maximizes the recovery of hidden value (Recall) without sacrificing overall predictive accuracy. Ultimately, this work helps bridge the gap between simple statistical forecasting and more practical, value-driven risk management in sports betting.

For example, Hvattum et al. proposed the incorporation of ELO ratings for the prediction of match outcomes in their study [24]. Also, Owramipur et al. suggested the usage of Baysian networks [25] to predict the outcomes for the Spanish League Barcelona team [26]. Similarly, Presetio et al. proposed the incorporation of logistic regression [27] to predict the outcomes of football matches [28]. Moreover, Martins et al. used a polynomial classifier for the prediction of football outcomes [29]. Schauberger et al. selected the method of random forests to predict the results of football matches in their study [30]. Various attributes of football players was studied by Danisk et al. in order to predict the result of football matches in their related paper [31]. Constantinou proposed a new machine learning model, named Dolores which is based on Hybrid Bayesian Networks, to predict football match outcomes from datasets selected in a series of countries [32]. Baboota et al. applied gradient boosting [33] for English Premier League [34]. Additionally, a deep learning framework was designed from Rahman to tackle the problem of prediction football results [35].

To address the limitations of individual algorithms and the common issue of probability distortion in sports forecasting, we propose a unified benchmarking and calibration framework. Our methodology focuses on three core contributions:

- A Consistent Feature Engineering Pipeline: We build a predictive feature vector that combines dynamic team performance indicators—such as Elo ratings [24], Home Clean Sheet, and Away Fail-To-Score ratios—with raw bookmaker market odds. These odds are systematically adjusted using the Newton–Raphson method to isolate the intrinsic probabilities and remove market margins.
- Comprehensive Architectural Benchmarking: We conduct a comparative evaluation of different learning paradigms under strictly uniform conditions. By testing models ranging from linear approaches (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree , LightGBM, Random Forest) and Neural Network models (MLP), we provide an unbiased look at how different models handle the high-dimensional volatility of football data.
- Framework for Robust Betting Signals: To bridge the gap between model scores and actionable betting decisions, we propose the integration of an Isotonic Calibration layer as part of this framework. While the present study focuses on establishing and benchmarking the underlying predictive models, this calibration layer — to be applied in the next research phase — is intended to transform raw output scores into mathematically objective probabilities, enabling the derivation of robust betting

signals. As empirical results demonstrate, this process highlights clear performance trends across both the Home Win and Away Win markets:

  - Performance Stability: Linear models and methods, such as Logistic Regression and Random Forest, provide high structural stability and consistently competitive Precision across both markets.
  - Non-Linear Interaction Mapping: The Multi-Layer Perceptron (MLP) acts as the most balanced model, consistently achieving the highest Recall in both markets. This suggests that the MLP is better at mapping complex feature interactions and identifying matches with higher expected betting value that traditional statistical models often miss.

- Reproducible Large-Scale Evaluation: Unlike many previous studies that evaluate a single predictive model or employ heterogeneous experimental settings, all algorithms are trained and evaluated using the same large-scale dataset, an identical feature space, and a chronological train–test split that better reflects real-world forecasting conditions while preventing information leakage. This unified evaluation protocol enables a fair comparison of fundamentally different machine learning paradigms and improves the reproducibility of the reported results.

Overall, the proposed framework provides a reproducible methodology for evaluating multiple machine learning paradigms under identical experimental conditions. The experimental results indicate that the MLP captures complex non-linear feature interactions more effectively than the remaining benchmarked methods, while the Random Forest exhibits superior precision and stability. These findings support the development of calibrated predictive systems for football match forecasting and subsequent probability-based decision making.

Although numerous studies have reported promising results for football match prediction, direct numerical comparison remains challenging because different datasets, leagues, feature sets, and evaluation protocols are typically employed. Rather than claiming a universally superior predictive accuracy, the present work focuses on providing a unified and reproducible benchmarking framework in which seven fundamentally different machine learning paradigms are evaluated under identical experimental conditions on a large-scale dataset. This allows a fair assessment of their relative strengths while emphasizing model stability and probability reliability through repeated experimentation and calibration.

The remaining of this article is organized as follows: in section 2 the proposed datasets and the used methodology are fully described, in section 3 the experimental results are outlined and finally in section 4 a discussion is provided based on the experimental results.

2. Materials and Methods

2.1. The Dataset Employed

In this study, we utilized a large-scale dataset of historical football matches sourced from a professional sports analytics repository. The data collection process prioritized high granularity, incorporating not only match results but also advanced performance indicators and pre-match market probabilities. The dataset's reliability is ensured through its internal consistency and widespread use in sports forecasting benchmarks.

2.2. Dataset Description

The experimental dataset comprises 225,474 match events spanning from 2018 to 2025. To ensure rigorous evaluation, the data was partitioned into a training set of 149,998 matches and a dedicated test set of 75,476 matches. The core variables included in the raw data are summarized in Table 1. The inclusion of *LeagueId* and *SeasonPhase* allows the model to account for competitive context and seasonal variations. To avoid information leakage and better reflect a real-world deployment scenario, the dataset was partitioned chronologically, with earlier seasons used for training and later matches reserved exclusively for testing.

Table 1. Description of the original dataset features.

Feature	Description / Range
MatchDate	Temporal identifier (2018–2025)
League / LeagueId	Categorical identifiers for competitive tiers
HomeTeam / AwayTeam	Competing squad identifiers
Label	Primary outcome target (1, X, 2)
EloDiff	Difference in Elo ratings (-500 to $+500$)
ProbHome/Draw/Away	Pre-match market probabilities (0 to 1)
RankDifference	Difference in league table standings
SeasonPhase	Categorical marker for season stage
AttackVsDefense / DefenseVsAttack	Derived offensive and defensive interaction ratios
TotalScoringPotential	Normalized long-term scoring capacity
TotalDefensiveWeakness	Normalized long-term goal concession rate
HomeCleanSheetRatio	Probabilistic historical clean sheet frequency
AwayFTSRatio	Away team Failed-to-Score probability

2.3. Feature Engineering and Mathematical Formulation

To capture the dynamic interaction between offensive and defensive capacities, the raw data was transformed into formal performance metrics. Let $AvgH_s$ and $AvgH_c$ denote the average goals scored and conceded by the home team, and $AvgA_s$, $AvgA_c$ the corresponding values for the away team. A smoothing constant $\epsilon = 0.01$ was applied to prevent division by zero.

AttackVsDefense: Represents the home team's offensive efficiency relative to the away team's defensive record:

$$\text{AttackVsDefense} = \frac{AvgH_s}{AvgA_c + \epsilon} \tag{1}$$

DefenseVsAttack: Quantifies the home team's defensive capability against the away team's scoring force:

$$\text{DefenseVsAttack} = \frac{AvgH_c}{AvgA_s + \epsilon} \tag{2}$$

TotalScoringPotential: Modeled as the numerical difference between home and away scoring capacities, identifying the relative offensive edge:

$$\text{TotalScoringPotential} = AvgH_s - AvgA_s \tag{3}$$

TotalDefensiveWeakness: Assesses the relative vulnerability of the home defense compared to the away defense as a ratio:

$$\text{TotalDefensiveWeakness} = \frac{AvgH_c}{AvgA_c + \epsilon} \tag{4}$$

HomeCleanSheetRatio (HCSR) and ***AwayFTSRatio*** (AFTS) capture probabilistic historical frequencies:

$$HCSR = \frac{\sum_{i=1}^{N} \mathbb{I}(G_{H,c} = 0)}{N}, \quad AFTS = \frac{\sum_{j=1}^{N} \mathbb{I}(G_{A,s} = 0)}{N} \tag{5}$$

2.4. Feature Importance Motivation

The generated feature space is explicitly designed to satisfy three essential, complementary operational criteria:

1. Long-term structural team performance using rating systems (*EloDiff*, *RankDifference*).
2. Granular tactical interaction effects (*AttackVsDefense*, *DefenseVsAttack*).
3. Distribution-based prior probabilities derived from betting market evaluations (*ProbHome*, *ProbDraw*, *ProbAway*). This multi-source configuration mitigates model vulnerability to single-feature bias and improves generalization accuracy.

3. Results

3.1. Experimental settings

For the development and training of the predictive models for sports outcomes, the ML.NET [36] framework was employed—an open-source, cross-platform machine learning framework developed by Microsoft. The choice of ML.NET was driven by its capability to integrate sophisticated algorithms, such as FastForest (a highly optimized implementation of the Random Forest), which offers both computational efficiency and robustness when handling large-scale datasets. To evaluate the stability of the models, a 30-run stability analysis was conducted. This rigorous validation process ensured consistent performance metrics (Error, Precision, and Recall) and minimized the risk of overfitting, thereby validating the model's reliability for analyzing noisy, real-world data. The following methods were incorporated in the conducted experiments from ML.NET:

1. FastTree [21].
2. Gam [20].
3. LightGBM [22].
4. Logistic Regression [27].
5. MLP, that denotes an artificial neural network [13,14] with 10 processing nodes. This network is trained using the L-BFGS method [37].
6. Random Forest [23].
7. RBF, that denotes a Radial Basis Function network [38,39].

The MLP was trained with 10 L-BFGS iterations (Table 2). To confirm this was sufficient for convergence, a control run with 100 iterations was also conducted; the resulting Precision, Recall, and Error values differed by no more than 1.1 percentage points from those reported in Tables 3 and 4 for both markets, confirming that the model had already converged within the original 10-iteration budget.

Table 2 presents the parameters used by the machine learning methods employed in the conducted experiments.

Table 2. Experimental settings.

METHOD	PARAMETER	VALUE
FASTTREE	Leaves	20
FASTTREE	Trees	600
FASTTREE	MinExample	100
FASTTREE	LearnRate	0.05
FASTTREE	FeatureFraction	0.9
RandomForest	Leaves	20
RandomForest	Trees	100
RandomForest	MinExample	10
RandomForest	FeatureFraction	0.8
LightGBM	Leaves	31
LightGBM	MinExample	20
LightGBM	LearnRate	0.1
LightGBM	Iterations	100
GAM	Iterations	100
GAM	MaxBinCount	255
GAM	LearnRate	0.1
RBF	Clusters (K-means)	2
RBF	Sigma (Gaussian)	1.4
RBF	MaxIter	200
MLP	LBFGS Iterations	10
Logistic Regression	L2Regularization	0.1
Logistic Regression	Tolerance	10^{-4}

These parameters are analyzed in detail below:

1. **Leaves**: Defines the complexity of each tree by setting the maximum number of leaves. A higher value allows the model to capture more complex patterns but increases the risk of overfitting.
2. **Trees**: Specifies the total number of trees to build. Increasing this number generally improves model stability and predictive performance but requires more training time.
3. **MinExample**:Sets the minimum number of data samples required to create a leaf. This acts as a regularization mechanism; higher values prevent the model from learning from noise or outliers.
4. **LearnRate**: Controls the contribution of each tree to the final prediction. A lower learning rate makes the optimization process more robust, requiring more trees to reach an optimal solution.
5. **FeatureFraction**: Determines the percentage of features to consider when building each tree. Using a fraction (e.g., 0.8) introduces randomness, which helps the model avoid focusing too heavily on a single feature and improves generalization.
6. **MaximumIterations** (Iterations): The number of times the algorithm passes through the entire training dataset to update weights.
7. **MaxBinCount**: Sets the maximum number of bins used to group features, helping to discretize continuous variables for more efficient processing.
8. **Clusters** (K-Means): Defines the number of centroids (hidden neurons) used to represent the input space. Each cluster captures a local region of the data.
9. **Sigma** (Gaussian Spread): Controls the width of the radial basis function (Gaussian kernel). It determines how far the influence of a single neuron extends in the input space; a smaller sigma makes neurons more sensitive to localized inputs.

10. **L2Regularization**: A penalty applied to the magnitude of the model weights. It discourages overly large weights, which helps in preventing overfitting and improves the model's ability to generalize to unseen data.
11. **Tolerance**: The threshold for the optimization algorithm to determine convergence. It dictates how small the improvements in the loss function must be before the algorithm stops.

3.2. Experimental results

Definition of the two binary problems. Before presenting the numerical results, it is important to clarify how the target variable was constructed for each market. The original dataset label (Label $\in$ {1, X, 2}) corresponds to the three-way outcome of a match (home win, draw, away win). For the purposes of this analysis, the problem was not treated as a three-class classification task, but was instead reformulated into two separate binary classification problems, one for each betting market:

Home Win market (Table 3): The positive class corresponds to the outcome "1" (home win), while the classes "X" (draw) and "2" (away win) were merged into a single negative class, "not Home Win." In other words, the model is asked to answer the question "will the home team win or not," collapsing the distinction between a draw and an away win.

Away Win market (Table 4): Conversely, the positive class corresponds to the outcome "2" (away win), while the classes "X" (draw) and "1" (home win) were merged into a single negative class, "not Away Win."

This approach was chosen because it directly mirrors how real-world betting markets of the "Double Chance" / single-market type are structured (e.g., "1" versus "X2," and "2" versus "1X"), and it allows each model to be independently optimized for the specific market of interest, without the constraints of a shared multi-class loss function. It is worth noting that this merging directly affects class balance: in the Away Win market, the positive class ("2") is the minority category in most leagues, which partly explains the systematically lower Recall values observed both in Table 4 and in the per-league stability analysis (§3.3, Figures 1–3).

Table 3 depicts the experimental results using the incorporated method on the Home Win dataset.

Table 3. Home win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	34.63%	64.91%	63.28%
GAM	34.89%	64.82%	62.75%
LightGBM	34.91%	64.59%	63.00%
Logistic Regression	34.51%	65.03%	63.44%
MLP	35.05%	64.21%	63.94%
Random Forest	34.71%	65.11%	62.86%
RBF	44.31%	54.75%	54.71%

Home Win results (Table 3). Table 3 summarizes the performance of the seven methods on the binary "Home Win vs. merged X2" problem. Random Forest achieved the highest Precision overall (65.11%), with Logistic Regression close behind (65.03%) and also recording the lowest Error among all seven methods (34.51%). Together, these two methods confirm their role as the most structurally stable, high-precision baselines for this market. In contrast, the MLP, despite showing the highest error (35.05%) and the lowest Precision (64.21%) in the group, stood out with the highest Recall (63.94%), indicating that the network recovers a larger proportion of actual home-win instances, even at the cost of somewhat lower precision on its positive predictions. RBF remained clearly the weakest model (Error 44.31%, Precision 54.75%, Recall 54.71%), confirming its limited suitability for this task under the current parameterization (Table 2).

Similarly, Table 4 presents the experimental results on the Away win dataset with the assistance of the the used machine learning methods.

Table 4. Away win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	27.97%	68.04%	59.29%
GAM	28.07%	68.64%	58.36%
LightGBM	27.96%	68.07%	59.31%
Logistic Regression	27.82%	68.21%	59.66%
MLP	27.90%	67.59%	60.25%
Random Forest	27.96%	69.07%	58.39%
RBF	32.62%	56.68%	52.91%

Away Win results (Table 4). In the binary "Away Win vs. merged X1" problem, absolute error levels are lower than in the Home Win market (Error 27.82%–28.07% for the six strongest methods), reflecting the fact that away wins occur less frequently overall, which mechanically lowers the baseline error rate for a model that leans conservative. Random Forest again delivered the highest Precision (69.07%), followed by GAM (68.64%) and Logistic Regression (68.21%), reinforcing the pattern observed in the Home Win market: tree-based and linear methods excel at producing high-confidence positive predictions. The MLP, once more, achieved the highest Recall (60.25%) while trailing slightly in Precision (67.59%) and Error (27.90% — still competitive, and in fact better than several tree-based alternatives). RBF again underperformed substantially across all three metrics (Error 32.62%, Precision 56.68%, Recall 52.91%), confirming its consistent weakness across both markets.

Overall pattern. Taken together, Table 3 and Table 4 reveal a consistent trade-off across both merged binary formulations: Random Forest and Logistic Regression prioritize Precision and stability, making fewer but more confident positive predictions, whereas the MLP prioritizes Recall, capturing a larger share of true positive outcomes (whether "Home Win" or "Away Win") at a modest cost in precision. This trade-off is directly relevant to betting-signal generation, since a higher-Recall model such as the MLP is more likely to surface value opportunities that higher-Precision models miss, while a higher-Precision model such as Random Forest is more likely to minimize false-positive betting signals. This distinction motivates the subsequent stability analysis in §3.3, where the behavior of the (higher-Precision) Random Forest model is examined in greater depth across leagues and repeated runs, and it also motivates the planned Isotonic Calibration layer, which aims to reconcile these Precision/Recall trade-offs into a single, well-calibrated probability estimate for each of the two merged binary markets.

3.3. Statistics

The following five figures were produced from the 30-run stability analysis and the overall model-comparison results contained in the accompanying experimental workbook. They are intended to supplement the experimental results with a visual, league-level and run-level view of model robustness, and to support the completed Conclusions section at the end of this document. The Figure 1 illustrates the average precision, recall, and classification error achieved by the Random Forest model (200 trees, averaged over 30 runs) for the Home Win and Away Win markets. The results are reported for both the pooled dataset ("All Leagues") and each of the ten leagues evaluated individually.

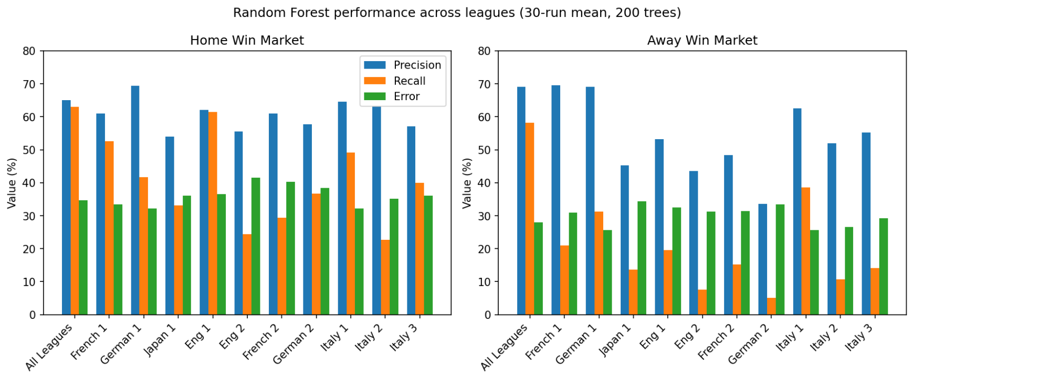

Figure 1. Mean Precision, Recall and Error (%) of the Random Forest model (200 trees, 30-run average) for the Home Win and Away Win markets, compared across the pooled dataset ("All Leagues") and the ten individually evaluated leagues.

The grouped comparison, now extended to all ten individually evaluated leagues alongside the pooled dataset, confirms that Precision remains comparatively stable across leagues (53.98% for Japan 1 to 69.36% for German 1, Home Win; 33.60% for German 2 to 69.52% for French 1, Away Win), while Recall is markedly more league-dependent. For the Away Win market, Recall collapses from 58.24% on the pooled dataset to as low as 5.13% (German 2) and 7.55% (Eng 2), and reaches at most 38.55% (Italy 1) among the individually evaluated leagues — still well below the pooled figure. Home Win shows the same pattern, with Recall ranging from 22.74% (Italy 2) up to 61.49% (Eng 1), the latter being the only individual league to approach the pooled model's 63.04%. Since Away Win is already the minority outcome class in most domestic leagues, restricting training and evaluation to a single, smaller league sharply reduces the number of positive examples available to the model, consistently pushing the classifier towards more conservative, higher-precision but lower-recall behaviour.

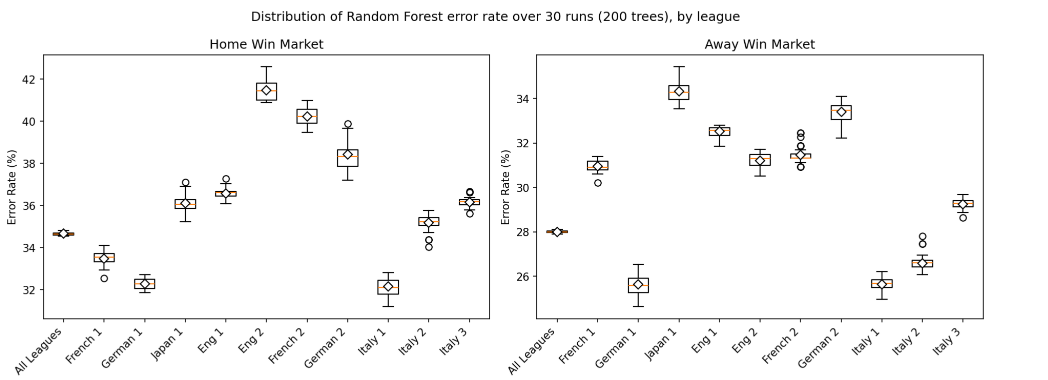

Figure 2. Distribution of the Random Forest Error Rate (%) over 30 independent runs (200 trees), shown separately for the Home Win and Away Win markets and for each league subset. Diamonds mark the mean; the horizontal line marks the median; circles denote outlier runs.

The box plot presented in Figure 2 is the clearest evidence of statistical stability in the study. The pooled ("All Leagues") boxes are visibly the narrowest for both markets, with an interquartile range of well under one percentage point, confirming that the model's error is not the product of a lucky or unlucky random seed. The single-league boxes are noticeably wider than the pooled model across all ten leagues, with the extremes illustrating the effect clearly: Home Win Error ranges from a low of 32.15% (Italy 1) to a high of 41.46% (Eng 2), while Away Win Error ranges from 25.62% (Italy 1) to 34.32% (Japan 1). Italy 1 is consistently the most stable individual league in both markets, while Eng 2 (Home Win) and Japan 1 (Away Win) show both the highest mean Error and the widest run-to-run spread — indicating that model performance on smaller, single-league samples is sensitive not only to Recall but to overall predictive stability, and more susceptible to the particular train/test partition drawn in each of the 30 runs. This supports training the calibrated model on the pooled, multi-league dataset rather than on isolated per-league subsets whenever a league-specific deployment is not strictly required.

Figure 3 displays the Precision–Recall results obtained from the 30 Random Forest runs for each league and market. The Home Win market is indicated by circles, while the Away Win market is represented by triangles.

Figure 3. Precision versus Recall for each of the 30 Random Forest runs, grouped by league and market (circles = Home Win, triangles = Away Win).

Each cluster of points corresponds to one of the twenty-two run-groups from Figure 1, and the tightness of a cluster is a direct visual proxy for run-to-run consistency. The pooled-dataset clusters (blue) are compact, confirming the low variance already observed in the box plot. The single-league clusters, by contrast, are visibly more elongated along the Recall axis — most clearly for French 1 Away Win (orange triangles), which occupies a narrow Precision band around 67–73% but a wide Recall band from roughly 19% to 23%. This pattern is consistent with the smaller effective sample size of single-league evaluation: Precision is estimated from a comparatively larger pool of predicted positives, while Recall depends on the (much smaller) count of true away-win matches actually present in that league's test fold, making it inherently noisier from run to run. This pattern generalizes across all ten leagues: German 2 Away Win shows the most extreme case, with Recall collapsing to a 30-run mean of just 5.13% despite a Precision of 33.60%, reflecting the very small number of true away-win matches available within that single league's test folds.

Figure 4 displays the classification error recorded over 30 Random Forest runs for the pooled ("All Leagues") model in the Home Win (top) and Away Win (bottom) markets. The dashed lines represent the corresponding mean error across all runs.

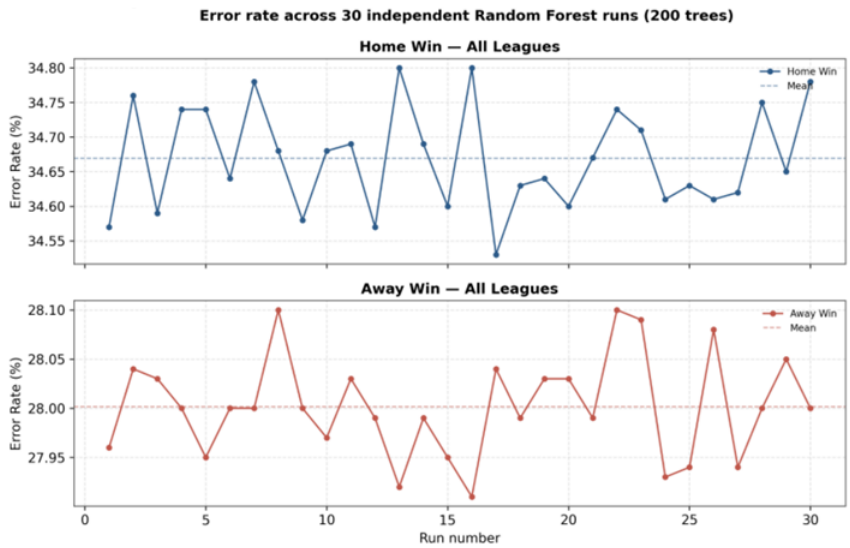

Figure 4. Error Rate (%) of the pooled ("All Leagues") Random Forest model across the 30 individual runs, for the Home Win (top) and Away Win (bottom) markets. Dashed lines mark the 30-run mean.

No individual run departs materially from the mean trend in either market: Home Win error oscillates within roughly 34.5–34.8% and Away Win error within roughly 27.9–28.1%, with no visible upward or downward drift across the sequence of runs. This absence of any isolated spike or systematic trend rules out a training-order or seed-dependent instability in the pooled configuration, and is consistent with the narrow boxes obtained for the same group in Figure 5.

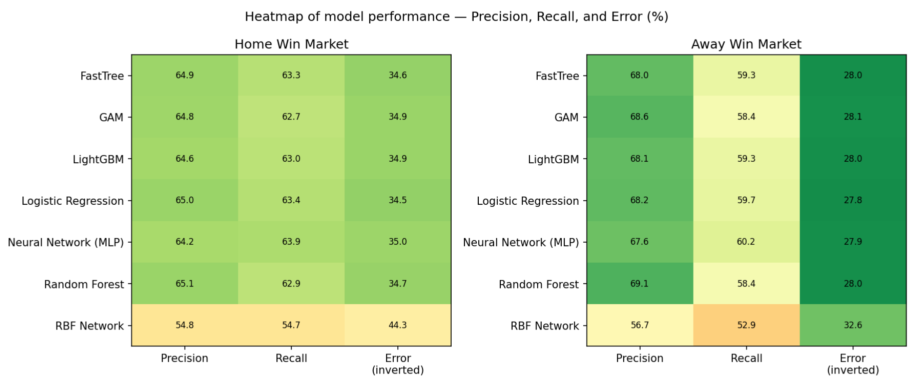

Figure 5. Heatmap of Precision, Recall and Error (%) for all seven benchmarked models on the Home Win (left) and Away Win (right) markets. The Error column is displayed inverted (100 − Error) so that green consistently denotes better performance across all three columns; the numeric labels show the original, non-inverted values.

The heatmap condenses Tables of the manuscript into a single, immediately readable panel. Random Forest and Logistic Regression stand out with the darkest green cells in the Precision column for both markets, confirming their role as the most structurally stable, high-precision methods. The MLP row is the greenest in the Recall column for both markets

(63.94% Home Win, 60.25% Away Win), visually reinforcing the paper's central claim that the neural network best recovers "hidden value" opportunities. RBF is uniformly the weakest model, appearing in red across nearly every cell for both markets, which visually isolates it as the clear outlier among the seven benchmarked algorithms.

4. Conclusions

This study presented a unified benchmarking and calibration framework for football match outcome prediction, evaluating seven machine learning paradigms — Logistic Regression, GAM, FastTree, LightGBM, Random Forest, RBF, and MLP — on a large-scale dataset of approximately 230,000 matches spanning 2018–2025. Under strictly uniform experimental conditions, Random Forest and Logistic Regression consistently delivered the highest Precision for both the Home Win (65.11% and 65.03%, respectively) and Away Win markets (69.07% and 68.21%, respectively), confirming their suitability as stable, low-variance baselines. The Multi-Layer Perceptron, in contrast, achieved the highest Recall in both markets (63.94% for Home Win and 60.25% for Away Win), indicating a superior ability to capture the non-linear feature interactions that drive less obvious, higher-value outcomes. RBF consistently underperformed across every metric and both markets, making it unsuitable for this task in its present configuration.

The 30-run stability analysis conducted on the Random Forest model reinforces these findings from a robustness perspective. When trained and evaluated on the full, pooled multi-league dataset, the model's Error Rate, Precision, and Recall all exhibited very low run-to-run variance (standard deviations below 0.3 percentage points for Error), confirming that the reported performance is a reliable structural property of the model rather than an artifact of a favourable random seed. However, when the same architecture was restricted to individual leagues — evaluated across all ten leagues available in the workbook — both the variance and the bias of the estimates increased substantially and consistently, most visibly in Recall for the Away Win market, which fell as low as 5.13% (German 2) and did not exceed 38.55% (Italy 1) in any single league, compared with 58.24% on the pooled data. Home Win Recall showed the same directional effect, though less severely, ranging from 22.74% (Italy 2) to 61.49% (Eng 1). This pattern held across every individually evaluated league without exception and is attributable to the reduced number of minority-class (away-win) examples available within a single league. It carries a practical implication for deployment: a pooled, multi-league training strategy should be preferred over per-league retraining whenever data volume in a target league is limited, in order to preserve both the stability and the recall of the resulting betting signals.

Taken together, these results show that combining rigorous multi-model benchmarking with an explicit stability analysis provides a more complete and trustworthy picture of predictive performance than single-run accuracy figures alone. As outlined in the Introduction, the next phase of this research applies the Newton–Raphson de-vigging procedure and an Isotonic Calibration layer to the raw outputs of these models, in order to transform them into objective, well-calibrated probabilities from which robust betting signals can be derived. While the present study substantially extends the league-level robustness analysis to ten individually evaluated leagues, future work will extend the present 30-run robustness protocol to the remaining six algorithms, to leagues beyond those currently available in the workbook, and will investigate whether league-aware feature engineering or transfer-learning strategies can recover the Recall lost when a model is confined to a single, smaller competition, without sacrificing the stability demonstrated by the pooled configuration.

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References

1. Raissi, M., & Karniadakis, G. E. (2018). Hidden physics models: Machine learning of nonlinear partial differential equations. Journal of Computational Physics, 357, 125-141.
2. Kashinath, K., Mustafa, M., Albert, A., Wu, J. L., Jiang, C., Esmaeilzadeh, S., ... & Prabhat, N. (2021). Physics-informed machine learning: case studies for weather and climate modelling. Philosophical Transactions of the Royal Society A, 379(2194), 20200093.
3. Viquar, M., Basak, S., Dasgupta, A., Agrawal, S., & Saha, S. (2019). Machine learning in astronomy: A case study in quasar-star classification. Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2018, Volume 3, 827-836.
4. Luo, S., Leung, A. P., Hui, C. Y., & Li, K. L. (2020). An investigation on the factors affecting machine learning classifications in gamma-ray astronomy. Monthly Notices of the Royal Astronomical Society, 492(4), 5377-5390.
5. Meuwly, M. (2021). Machine learning for chemical reactions. Chemical Reviews, 121(16), 10218-10239.
6. J.A. Aguiar, M.L. Gong, T.Tasdizen, Crystallographic prediction from diffraction and chemistry data for higher throughput classification using machine learning, Computational Materials Science **173**, 109409, 2020.
7. S.S. Yadav, S.M. Jadhav, Deep convolutional neural network based medical image classification for disease diagnosis. J Big Data **6**, 113, 2019.
8. L. Qing, W. Linhong , D. Xuehai, A Novel Neural Network-Based Method for Medical Text Classification, Future Internet **11**, 255, 2019.
9. Athey, S. (2018). The impact of machine learning on economics. In The economics of artificial intelligence: An agenda (pp. 507-547). University of Chicago Press.
10. R. Hafezi, J. Shahrabi, E. Hadavandi, A bat-neural network multi-agent system (BNNMAS) for stock price prediction: Case study of DAX stock price, Applied Soft Computing **29**, pp. 196-210, 2015.
11. Ghai, D., Tripathi, S. L., Saxena, S., Chanda, M., & Alazab, M. (Eds.). (2022). Machine learning algorithms for signal and image processing. John Wiley & Sons.
12. Ahmed, N. K., Atiya, A. F., Gayar, N. E., & El-Shishiny, H. (2010). An empirical comparison of machine learning models for time series forecasting. Econometric reviews, 29(5-6), 594-621.
13. Zou, J., Han, Y., & So, S. S. (2009). Overview of artificial neural networks. Artificial neural networks: methods and applications, 14-22.
14. Abiodun, O. I., Jantan, A., Omolara, A. E., Dada, K. V., Mohamed, N. A., & Arshad, H. (2018). State-of-the-art in artificial neural network applications: A survey. Heliyon, 4(11).
15. De Ville, B. (2013). Decision trees. Wiley Interdisciplinary Reviews: Computational Statistics, 5(6), 448-455.
16. Kotsiantis, S. B. (2013). Decision trees: a recent overview. Artificial Intelligence Review, 39(4), 261-283.
17. Pisner, D. A., & Schnyer, D. M. (2020). Support vector machine. In Machine learning (pp. 101-121). Academic Press.
18. Valkenborg, D., Rousseau, A. J., Geubbelmans, M., & Burzykowski, T. (2023). Support vector machines. American journal of orthodontics and dentofacial orthopedics, 164(5), 754-757.
19. Clarke, S. (2016). A review of methods for deriving probability estimates from betting odds.
20. Wood, S. N. (2017). Generalized Additive Models: An Introduction with R (2nd ed.). Chapman and Hall/CRC.
21. Price, M. N., Dehal, P. S., & Arkin, A. P. (2010). FastTree 2–approximately maximum-likelihood trees for large alignments. PloS one, 5(3), e9490.
22. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. Advances in neural information processing systems, 30.
23. Liaw, A., & Wiener, M. (2002). Classification and regression by randomForest. R News, 2(3), 18-22.s

24. Hvattum, L. M., & Arntzen, H. (2010). Using ELO ratings for match result prediction in association football. International Journal of forecasting, 26(3), 460-470.
25. Marcot, B. G., & Penman, T. D. (2019). Advances in Bayesian network modelling: Integration of modelling technologies. Environmental modelling & software, 111, 386-393.
26. Owramipur, F., Eskandarian, P., & Mozneb, F. S. (2013). Football result prediction with Bayesian network in Spanish League-Barcelona team. International Journal of Computer Theory and Engineering, 5(5), 812.
27. Stoltzfus, J. C. (2011). Logistic regression: a brief primer. Academic emergency medicine, 18(10), 1099-1104.
28. D. Prasetio and D. Harlili, "Predicting football match results with logistic regression," 2016 International Conference On Advanced Informatics: Concepts, Theory And Application (ICAICTA), Penang, Malaysia, 2016, pp. 1-5, doi: 10.1109/ICAICTA.2016.7803111.
29. Martins, R. G., Martins, A. S., Neves, L. A., Lima, L. V., Flores, E. L., & do Nascimento, M. Z. (2017). Exploring polynomial classifier to predict match results in football championships. Expert Systems with Applications, 83, 79-93.
30. Schauberger, G., & Groll, A. (2018). Predicting matches in international football tournaments with random forests. Statistical Modelling, 18(5-6), 460-482.
31. Danisik, N., Lacko, P., & Farkas, M. (2018, August). Football match prediction using players attributes. In 2018 World symposium on digital intelligence for systems and machines (DISA) (pp. 201-206). IEEE.
32. Constantinou, A. C. (2019). Dolores: a model that predicts football match outcomes from all over the world. Machine learning, 108(1), 49-75.
33. Natekin, A., & Knoll, A. (2013). Gradient boosting machines, a tutorial. Frontiers in neurorobotics, 7, 63623.
34. Baboota, R., & Kaur, H. (2019). Predictive analysis and modelling football results using machine learning approach for English Premier League. International Journal of Forecasting, 35(2), 741-755.
35. Rahman, M. A. (2020). A deep learning framework for football match prediction. SN Applied Sciences, 2(2), 165.
36. Microsoft. (2026). ML.NET: Machine Learning for .NET. Retrieved June 30, 2026, from https://dotnet.microsoft.com/en-us/apps/machinelearning-ai/ml-dotnet
37. Liu, D. C., & Nocedal, J. (1989). On the limited memory BFGS method for large scale optimization. Mathematical programming, 45(1), 503-528.
38. J. Park and I. W. Sandberg, Universal Approximation Using Radial-Basis-Function Networks, Neural Computation **3**, pp. 246-257, 1991.
39. G.A. Montazer, D. Giveki, M. Karami, H. Rastegar, Radial basis function neural networks: A review. Comput. Rev. J **1**, pp. 52-74, 2018.